

Harshit Rana

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PROFILE SUMMARY

Aspiring Software Engineer with a strong foundation in Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, and Computer Networks. Proficient in Java, Python, Flask, MySQL, PostgreSQL, REST APIs, and Full-Stack Development, with hands-on experience building scalable web applications and intelligent software solutions. Developed projects in civic issue management, developer productivity, and smart agriculture using modern software engineering practices. Passionate about designing efficient, scalable, and user-centric applications while continuously learning emerging technologies.

Education

Bachelor of Technology (Artificial Intelligence & Machine Learning) Aug 2023 – Present

Chandigarh Group of Colleges, Jhanjeri, Mohali

CGPA: 8.2

Senior Secondary Education (Class XII)

2023

St.Ezra International School, Punjab

Percentage: 65%

Secondary Education (Class X)

2022

Delhi Convent School, HP

Percentage: 86%

Internship Experience

Python Intern

Sep 2025 – Oct 2025

Oasis Infobyte 🌐

Remote

- Developed Python-based applications using Object-Oriented Programming (OOP), file handling, exception handling, and modular programming principles.
- Implemented debugging, testing, and code optimization techniques to improve application reliability and maintainability.
- Built multiple Python projects utilizing functions, loops, conditional statements, and data structures to solve practical programming problems.

Projects

Civic Issue Management Platform

🌐 GitHub Repository

- Developed a full-stack civic issue management platform for reporting, tracking, and resolving public infrastructure complaints using interactive GIS-based maps.
- Implemented role-based access control, user and administrator dashboards, complaint lifecycle management, real-time updates, and advanced search and filtering.
- Designed and developed REST APIs using Flask, PostgreSQL, and PostGIS for user authentication, complaint management, geospatial data processing, and analytics.
- Built a responsive web application integrating Leaflet.js for geospatial visualization, complaint status tracking, analytics dashboards, and report export functionality.

Developer Growth Platform

🌐 GitHub Repository

- Developed a personalized placement preparation platform enabling students to assess skills, track learning progress, and manage placement preparation resources.

- Designed and implemented modules for resume analysis, coding progress tracking, study planning, interview preparation, and placement readiness dashboards.
- Integrated LLM APIs to generate personalized learning roadmaps, project recommendations, interview guidance, and career insights based on user profiles.
- Built a scalable backend with REST APIs, user authentication, progress tracking, and responsive dashboards following modular software architecture.

Smart Agriculture Management Platform

GitHub Repository

- Developed a smart agriculture platform for crop prediction, farm analytics, and resource optimization using machine learning and data analysis techniques.
- Performed data preprocessing, feature engineering, model training, evaluation, and visualization to generate actionable agricultural insights.
- Implemented Prim's and Kruskal's graph algorithms to optimize irrigation network design and improve resource utilization.
- Built a modular application with interactive dashboards, predictive analytics, and decision-support features for sustainable farm management.

Technical Skills

Programming: Python, Java, C++, C

Web: HTML, CSS, JavaScript, REST APIs

Machine Learning: Exploratory Data Analysis, Regression, Classification, Model Evaluation

Libraries: Pandas, NumPy, Scikit-learn, Matplotlib

Frameworks: Streamlit, Flask, FastAPI

Databases: MySQL, MongoDB

Statistical Tools: R Programming, Tableau

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Tools: Git, GitHub, Jupyter Notebook, VS Code

Certifications

- AWS Academy Graduate – Generative AI Foundations
- Oracle Academy – Database Programming with PL/SQL
- Data Analytics Job Simulation – Deloitte (Forage)
- Python (Basic) Certification – HackerRank
- Introduction to Generative AI Studio – Google Cloud & Simplilearn

Achievements & Participation

- Selected as an Exhibitor at the AI Impact Summit 2026 (Pre-Summit Event), showcasing AI-based solutions and interacting with industry professionals.
- Successfully completed the 100 Days of Coding Challenge, demonstrating consistency in programming and strengthening problem-solving skills.
- Presented a technical project at a College-Level Project Exhibition, explaining the system architecture, implementation, and real-world applications to faculty and peers.